MFAE-YOLO: Multifeature Attention-Enhanced Network for Remote Sensing Images Object Detection

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***Abstract*— Object detection in aerial remote sensing images is essential for applications such as traffic management, public security, and ecological monitoring. However, existing methods struggle to handle multiscale objects, complex backgrounds, and frequent occlusions, resulting in inadequate global feature extraction and unstable multiscale loss calculations. To address these challenges, we propose the multifeature attention-enhanced you only look once (MFAE-YOLO) network. Our approach introduces a global feature fusion processing (GFFP) module to enhance global feature extraction. The backbone network incorporates fusion of channel, pixel, and spatial (FCPS) attention modules to strengthen feature representation, while C2F-feature pool extraction units (C2F-FPEUs) optimize pooling-based feature extraction. In addition, we propose an accurate intersection over union (AIoU) loss function to refine bounding box regression. Experiments on the NWPU VHR-10, RSOD, and DIOR datasets demonstrate that MFAE-YOLO surpasses state-of-the-art methods, achieving mAP_{50} values of 94.7%, 94.8%, and 67.0%, respectively. Ablation studies further validate the contributions of each module. The code is available at https://github.com/yiboCode/MFAE-YOLO**

***Index Terms*— Aerial remote sensing, attention mechanism, multiscale feature extraction, object detection.**

I. Introduction

With the rapid advancement of aerial remote sensing technology, the volume of captured and stored images has grown exponentially [1]. For example, the WorldView-4 satellite captures approximately 680 000 km^2 of imagery daily, necessitating efficient processing [2]. To fully use these vast datasets, extensive research has been undertaken across various applications. Among these, object detection is crucial, as deep learning techniques automatically identify, classify, and annotate objects in aerial remote sensing images [3]. This technology supports essential tasks such as traffic monitoring, natural disaster assessment, and urban construction planning [4].

Object detection in aerial remote sensing images presents unique challenges not commonly encountered in conventional image detection [5], [6]. These challenges can be categorized into three main areas.

1) *Scale Variation:* Objects in aerial images vary significantly in scale due to the extensive coverage by satellites and drones, encompassing entities such as buildings, vehicles, and pedestrians. Adapting networks to handle these scale variations remains challenging [7].
2) *Small Object Detection:* Aerial images frequently contain numerous small objects resulting from large capture distances [8]. With limited pixels and semantic information, these objects are difficult to detect and are often missed, despite their critical importance in analysis.
3) *Image Quality and Background Complexity:* Atmospheric conditions, lighting, and viewing angles can degrade image quality. Moreover, complex backgrounds, including terrain and forests, introduce substantial noise that complicates object detection. As illustrated in Fig. 1, complex backgrounds and diverse target scales pose significant challenges for accurate detection and labeling.

Recently, convolutional neural networks (CNNs) have driven significant advancements in remote sensing object detection [9]. These methods are primarily categorized into two-stage and one-stage algorithms. Two-stage algorithms, such as the region-CNN (R-CNN) family [10], [11], [12], [13], [14], first generate candidate regions, and then refine and classify them to achieve higher accuracy. In contrast, one-stage algorithms such as the you only look once (YOLO) family [15], [16], [17], [18], [19] and single shot multibox detector (SSD) [20] simultaneously classify and locate objects without generating candidate regions, enabling faster and more efficient real-time detection [21]. However, while CNNs are adept at extracting local features, they often struggle to capture global features, leading to missed detections in images with multiscale objects and complex backgrounds [22].

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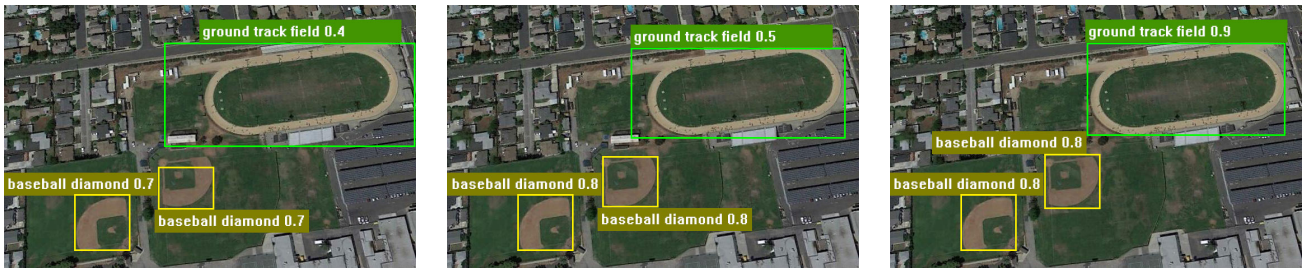
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(a) (b) (c)

Fig. 1. Comparison of object detection results in remote sensing images using different models. Our model demonstrates accurate target box localization in complex backgrounds with multiscale objects, as illustrated in the following results. (a) YOLOv5. (b) YOLOv8. (c) MFAE-YOLO.

Attention mechanisms have become prominent in image feature extraction [23], [24], [25], [26]. Notable models include Vision Transformers [27] and Swin Transformers [28]. Although these mechanisms excel at capturing global feature dependencies, they may compromise local feature details [29]. In remote sensing images, the similarity between global features of objects and their backgrounds can result in omissions and false detections in transformer-based models.

To address the limitations of purely convolutional or attention-based approaches, researchers have developed hybrid architectures that integrate both [30], [31], [32], [33], [34]. Peng et al. [35] introduced the Conformer, a network that combines convolution operations and self-attention to enhance learning. Li et al. [34] developed a global and local attention mechanism that fuses fine-grained local features with global background features, enabling comprehensive modeling of input images. These hybrid architectures have demonstrated strong performance on multiscale objects and complex backgrounds, underscoring their potential in remote sensing object detection.

Despite these advancements, the existing hybrid architectures still struggle to fully integrate local and global features. We propose that fusing these features in the initial stages of the model can enhance detection performance. This article introduces the multifeature attention-enhanced YOLO (MFAE-YOLO) network, tailored for aerial remote sensing images. Our approach optimizes the backbone layer with a hybrid architecture during initial image processing, thereby enhancing feature extraction. Furthermore, to improve the comprehensiveness and stability of feature fusion, we design a multilevel residual connection network that incorporates channel, pixel, and spatial attention (SA) mechanisms. This structure facilitates the integration of multiple global attention mechanisms with local feature extraction while also introducing more accurate shape loss calculations to accommodate the multiscale nature of objects.

The key contributions of this article are fivefold.

- An MFAE-YOLO network is designed for remote sensing object detection. To the best of our knowledge, this is the first work to construct a multifeature attention mechanism in YOLO's initial feature extraction and to optimize the subsequent convolution mechanism for effective integration. Its effectiveness is validated on multiple remote sensing datasets.
- A global feature fusion processing (GFFP) module that integrates multiple attention mechanisms and residual connections is proposed to stabilize global feature extraction.
- A fusion of channel, pixel, and spatial (FCPS) attention module is proposed, combining channel attention (CA), pixel attention (PA), and SA mechanisms to enhance multilevel feature representation.
- A C2F-feature pool extraction unit (C2F-FPEU) module is introduced to improve the original C2F module through adaptive feature pooling, specifically tailored for remote sensing images.
- An accurate intersection over union (AIoU) loss function is developed to enhance bounding box shape loss calculations, addressing limitations of the existing loss functions for bounding boxes with identical aspect ratios but varying widths and heights.

II. Related Works

A. Optimization Strategy for Remote Sensing Image Object Detection

Recent advances in remote sensing image object detection emphasize strategies such as image pyramids, feature pyramids, spatial pyramids, and loss function optimization [36]. The image pyramid strategy generates images at multiple scales through smoothing and downsampling, while the feature pyramid strategy connects feature maps of varying depths across neural network layers to enhance detection [37]. The spatial pyramid strategy uses parallel branches with different receptive fields to form a spatial pyramid, thereby improving object detection performance.

Object detection methods are generally divided into two-stage and one-stage algorithms. The one-stage methods estimate object locations and classify their categories in a single computation, which enables rapid detection but may lead to high false detection rates, particularly for small targets. In contrast, two-stage methods use two independent network models to achieve high accuracy, especially in scenarios involving large targets and complex scenes, albeit at a slower speed. Both the algorithm types incorporate these feature extraction strategies. Notable examples of two-stage algorithms include Faster R-CNN [38], Mask R-CNN [39], and Cascade R-CNN [40], which have evolved from the R-CNN framework.

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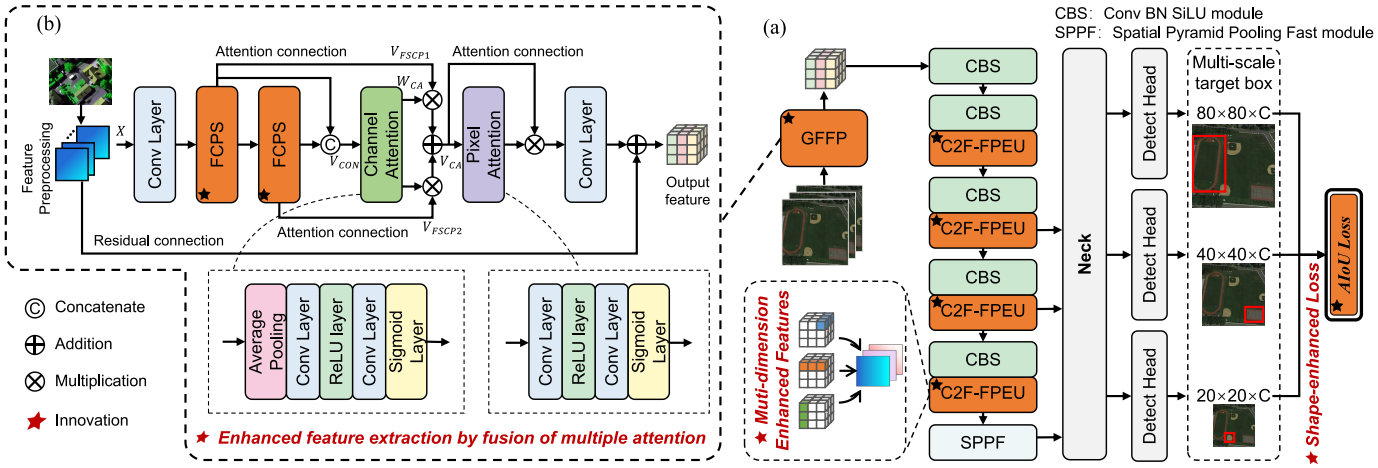


Fig. 2. Architecture of MFAE-YOLO highlighting three key innovations: 1) enhanced global feature extraction through the integration of multiple attention mechanisms, 2) improved multidimensional feature extraction, and 3) shape feature enhancement in loss function computation. (a) Overview of the MFAE-YOLO. (b) GFFP module.

The YOLO series [19], [41], [42], [43], [44], belonging to the one-stage category, uses both the feature and spatial pyramid strategies to accurately extract multiscale features, making it widely used in remote sensing image object detection. Therefore, this article focuses on further improvements in this area.

B. YOLO Enhancements

Given the multiscale nature of remote sensing images, the presence of small objects, and complex backgrounds, researchers have proposed numerous enhancements to YOLO, including high-performance modules and adaptive network structures. Several representative studies in recent years include: Li et al. [45] introduced CSP1-GhostnetV2 and VOVCSP-GhostnetV2 to improve YOLOv5 by creating a lightweight feature aggregation network; Tong and Wu [46] developed RMCS and MFEC components to enhance YOLOv8 for detecting small objects in large-scale images; and Wang et al. [47] combined ShuffleNet v2 with YOLOv5n to create an ultralightweight network for aerial remote sensing. Other improvements include a dual-branch attention mechanism [48], which integrates a visual transformer module to enhance the detection of small objects by modifying YOLOv8 and introducing the RIoU loss function, as well as the lightweight plug-and-play modules [49], resulting in the efficient small object detector FFCA-YOLO.

Currently, many enhancements target the neck or head layers; however, there remain shortcomings in optimizing the backbone for initial feature extraction to improve detection performance across different object sizes in remote sensing images. Moreover, the utilization of attention mechanisms needs further improvement. This article focuses on enhancing YOLO by improving both local and global feature extraction and optimizing the loss function.

C. Optimization of Loss Functions

The intersection over union (IoU) loss function is commonly used in object detection to measure the ratio between the intersection and union of predicted and annotated boxes. In recent years, researchers have introduced several improvements to the IoU loss function. Generalized IoU (GIoU) [50] addresses the limitations of IoU for nonoverlapping boxes, while distance IoU (DIoU) [51] incorporates the distance between bounding box centers. Complete IoU (CIoU) further improves performance by considering scale and aspect ratio [52]. Lin et al. [53] introduced the focal loss function to enhance dense small object detection, and Zhang et al. [54] replaced CIoU with focal IoU loss to accelerate convergence in object detection.

Although several studies have focused on the calculation methods of object detection loss values, especially regarding the proportions of detection boxes and the categories of detection, research on distance loss, angle loss, and shape loss remains limited. This article proposes a novel shape loss calculation method to address the limitations of CIoU, particularly when bounding boxes share the same aspect ratio but differ in width and height.

III. Proposed Method

A. Overview

As illustrated in Fig. 2(a), the MFAE-YOLO architecture comprises three main components: backbone, neck, and head. The backbone, serving as the network's core, processes input image data through the GFFP module to enhance global feature extraction. Within the backbone, the C2F-FPEU module effectively fuses global and local features at various stages using attention mechanisms and CNNs, thereby improving detection accuracy and generalization. This process yields feature maps rich in semantic information while preserving high spatial resolution.

Features are downsampled in the backbone via conv-batch normalization-sigmoid (CBS) modules and subsequently upsampled in the neck for further refinement. In the head, we introduce a new loss calculation method, AIoU, which selects features at three different scales and computes loss values for each, ensuring effective extraction of multiscale object information in remote sensing images.

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B. GFFP Module

The GFFP module, shown in Fig. 2(b), plays a pivotal role in the MFAE-YOLO network. It comprises two FCPS modules, one PA module, and one CA module, all integrated with multiple residual connections to ensure stable feature extraction.

Within the MFAE-YOLO framework, the GFFP module initiates global feature extraction as remote sensing images enter the network. By emphasizing global features early, the GFFP module establishes a robust foundation for efficient and effective feature extraction in subsequent layers.

1) Preliminary Feature Extraction and Merging: In the GFFP module, input remote sensing image data are first preprocessed using YOLO to generate a digital feature map, x, as depicted in Fig. 2(b). Two FCPS modules are then applied for multifeature attention fusion. The outputs of these modules are concatenated and merged to produce the extracted feature representation, V_{CON}

$$V_{\mathrm{CON}} = \mathrm{Concat}\begin{pmatrix} \mathrm{FCPS}(\mathrm{Conv}(x)) \\ \mathrm{FCPS}(\mathrm{FCPS}(\mathrm{Conv}(x))) \end{pmatrix}. \tag{1}$$

2) Feature Depth Extraction and Attention Connection: Following the initial merging, the feature representation V_{CON} undergoes channel-level feature extraction using the FCA function, as shown in (2) and (3). The FCA function combines outputs from the two FCPS modules to extract channel features and generate CA weights, W_{CA}. These weights are applied to the FCPS outputs, and the results are summed to optimize channel-level feature extraction

$$W_{\mathrm{CA}} = F_{\mathrm{CA}}(V_{\mathrm{CON}}) \tag{2}$$

$$V_{\mathrm{CA}} = V_{\mathrm{FCPS1}} \times W_{\mathrm{CA}} + V_{\mathrm{FCPS2}} \times W_{\mathrm{CA}}. \tag{3}$$

The resulting V_{CA} is then processed for pixel-level feature extraction using the FPA function. PA weights derived from V_{CA} are multiplied by V_{CA} to extract pixel features. The final feature map is computed by convolving the product with $C \times 3 \times 3$ kernels and adding the initial feature map x, thereby ensuring stable feature extraction

$$F_{\mathrm{output}} = X + \mathrm{Conv}^{C\times 3\times 3}(V_{\mathrm{CA}} \times F_{\mathrm{PA}}(V_{\mathrm{CA}})). \tag{4}$$

C. FCPS Module

Due to the complexity of the GFFP module, its internal FCPS module has been provided with additional nomenclature and explanation. The FCPS module is a crucial component of the GFFP module, integrating SA, CA, and PA mechanisms to facilitate multilevel feature extraction. This combination significantly enhances the network's ability to extract comprehensive features from image data. The FCPS module is shown in Fig. 3. The FCPS module consists of three sequential layers: CA, PA, and SA. Detailed calculations for each layer are provided below.

1) CA Layer: The CA layer processes a feature map x with dimensions (C, H, W), where C is the number of channels, H is the height, and W is the width. First, average pooling is applied along the spatial dimensions, yielding a tensor of size $(C, 1, 1)$. This tensor is then convolved with a 1×1 kernel followed by a rectified linear unit (ReLU) activation to downsample the features

$$f_{\mathrm{CAL}}(x) = \mathrm{ReLU}\big(\mathrm{Conv}^{1\times 1\times 1}(\mathrm{AvgPool2D}(x))\big). \tag{5}$$

Next, the features are upsampled using $C1 \times 1$ convolution kernels, normalized with a sigmoid function, and multiplied by the initial input x to produce the CA layer output

$$y = \tilde{f}_{\mathrm{CAL}}(x) = x \cdot \sigma\big(\mathrm{Conv}^{C\times 1\times 1}(f_{\mathrm{CAL}}(x))\big) \tag{6}$$

where σ denotes the sigmoid activation function.

2) PA Layer: The PA layer processes the feature map y with dimensions (C, H, W). At the pixel level, convolution operations are performed using $C/T\,1 \times 1$ convolution kernels (with $C \geq T$) and a stride of 1, extracting pixel-level features to yield a feature map of size $(C/T, H, W)$. A ReLU activation function introduces nonlinearity

$$f_{\mathrm{PAL}}(y) = \mathrm{ReLU}\big(\mathrm{Conv}^{C/T\times 1\times 1}(y)\big). \tag{7}$$

Subsequently, a 1×1 convolution further extracts depth features, producing a feature map of size $(1, H, W)$. This map is normalized using a sigmoid function and multiplied by the input y, followed by a residual connection to generate the PA layer output

$$z = \tilde{f}_{\mathrm{PAL}}(y) = y \cdot \sigma\big(\mathrm{Conv}^{1\times 1}(f_{\mathrm{PAL}}(y))\big). \tag{8}$$

3) SA Layer: In the SA layer, the feature map z with dimensions (C, H, W) is processed as follows:

$$f_{\mathrm{SAL}}(z) = z \cdot \sigma(\mathrm{Conv}(\mathrm{Concat}(\mathrm{Avg}(z), \mathrm{Max}(z)))). \tag{9}$$

In the SA layer, both average pooling and max pooling are applied along the channel dimension of z, yielding two feature maps of size $(1, H, W)$. These maps are concatenated to form a tensor of size $(2, H, W)$, which is then processed by $C1 \times 1$ convolution kernels to extract spatial features while preserving the original dimensions. The resulting tensor is normalized using a sigmoid activation function and multiplied by the input feature map z, followed by a residual connection to ensure stable feature extraction.

In the existing methods, the convolutional block attention module (CBAM) shares similarities with the proposed FCPS module, as CBAM primarily integrates channel and SA mechanisms sequentially. In contrast, the FCPS module introduces a novel approach by combining channel, pixel, and SA mechanisms after data preprocessing. Specifically, the CA mechanism compresses the spatial dimension to focus on extracting meaningful information, while the SA mechanism compresses the channel dimension to emphasize the positional information of the target. In addition, the PA mechanism integrates multiscale feature maps in remote sensing tasks, learning the importance of pixels at different scales to achieve more effective feature fusion. This integrated approach enables the FCPS module to capture both local and global contextual information more effectively, potentially improving performance in complex scenarios.

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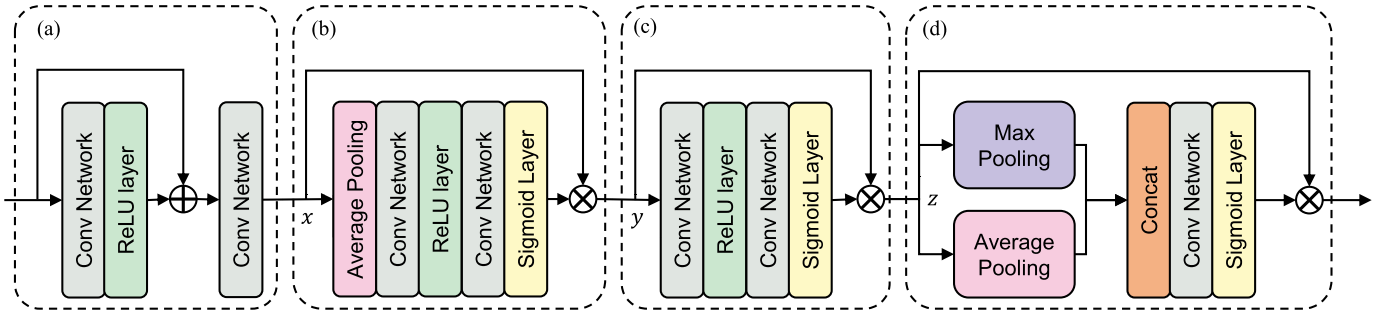


Fig. 3. FCPS attention module. After preprocessing, the CA mechanism, PA mechanism, and SA mechanism are combined in serial order. (a) Feature preprocessing. (b) CA layer. (c) PA layer. (d) SA layer.

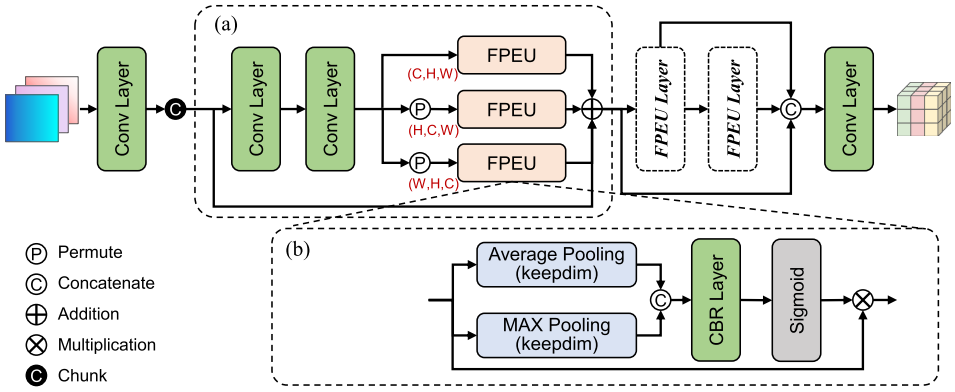


Fig. 4. Detailed illustration of the C2F-FPEU module. The module consists of three consecutive FPEU layers, each FPEU layer containing three FPEU modules. The FPEU module uses the pooling method for feature enhancement and finally merges the results of the three FPEU layers. (a) FPEU layer. (b) FPEU module.

D. C2F-FPEU Module

The C2F-FPEU module, shown in Fig. 4, is specifically designed to enhance feature extraction for remote sensing images. It incorporates three feature pooling extraction units (FPEU) that adaptively extend the original C2F module by leveraging attention mechanisms that combine max pooling and mean pooling. This combination improves the module's ability to capture relevant features effectively, as validated by ablation experiments.

The C2F-FPEU module comprises two convolutional layers, one Chunk layer, three FPEU layers, and one concatenation layer. The primary enhancement lies in the modified FPEU structure relative to the original C2F module. As shown in Fig. 4(b), the core FPEU module calculation is represented by

$$F = \sum(\sigma(\mathrm{CBR}(\mathrm{Concat}(\mathrm{Avg}(x), \mathrm{Max}(x))))). \tag{10}$$

In the C2F-FPEU design, the number of bottleneck layers, n, ranges from 1 to 3. The maximum and average values are extracted while preserving dimensionality for merging and subsequent dimensionality reduction. The CBR module comprises convolutional layers, batch normalization, and ReLU activation. The final output of the FPEU is the cumulative sum of the outputs from all three FPEUs.

Among the existing C2F improvement methods, the C2F-RFA module shares similarities with the proposed C2F-FPEU module in that use global average pooling or global max pooling to capture global information. However, the C2F-RFA module introduces additional interactions with receptive field feature information to learn attention maps, incurring extra computational overhead. In contrast, the proposed C2F-FPEU module retains the use of global average or max pooling to extract global information and uses 3-D transformations to better capture pooling effects from different dimensions before pooling, thereby reducing model complexity while maintaining effective feature extraction. This design achieves a reduction in computational overhead without compromising performance.

E. AIoU Loss Function

YOLOv8 uses the CIoU loss function to compute object detection loss, as described below. The CIoU loss function incorporates two additional components: distance loss and shape loss

$$\mathcal{L}_{\mathrm{CIoU}} = 1 - \frac{B_{\mathrm{gt}} \cap B_{\mathrm{prd}}}{B_{\mathrm{gt}} \cup B_{\mathrm{prd}}} + \frac{d}{c} + \mathrm{a\ v} \tag{11}$$

$$d = \rho^2\left(B_{\mathrm{gt}}, B_{\mathrm{prd}}\right) \tag{12}$$

$$c = (w_c)^2 + (h_c)^2 + \epsilon \tag{13}$$

where d is the squared distance between the center points of the predicted and ground-truth boxes, and c represents the squared diagonal distance of the smallest enclosing rectangle covering both the boxes. The term d/c introduces distance loss into the overall computation, while ϵ, typically set to 1×10^{-7},

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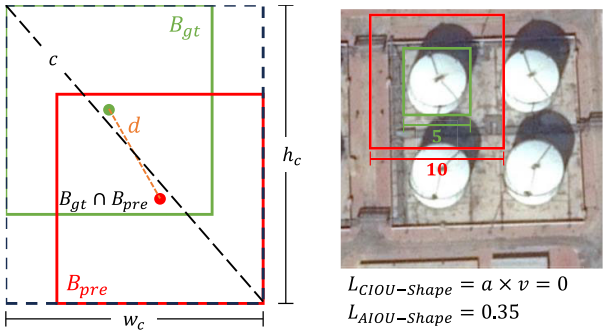


Fig. 5. Factors affecting box loss calculation in object detection models and our improvements. (a) Elements in IoU and distance loss, where the red box represents the model prediction box and the green box represents the actual box of the object.(b) Shape loss in CIoU and AIoU. When the center points of the target box and the predicted box overlap but have different sizes, the loss value calculation of CIOU becomes invalid, while our proposed AIOU method can obtain the shape loss value.

prevents division by zero

$$a = \frac{v}{(1 - \mathrm{IoU}) + v} \tag{14}$$

$$v = \frac{4\left(\arctan \frac{w_{\mathrm{gt}}}{h_{\mathrm{gt}}} - \arctan \frac{w_{\mathrm{prd}}}{h_{\mathrm{prd}}}\right)^2}{\pi^2} \tag{15}$$

where v refines the shape loss by considering the difference in aspect ratios between the predicted and ground-truth boxes. However, when the predicted and ground-truth boxes share the same aspect ratio but differ in size, v becomes zero, rendering the shape loss ineffective.

To address this limitation, we propose the AIoU loss function. AIoU incorporates both the shape and size of objects into the shape loss calculation, making it particularly effective for remote sensing images containing large objects with diverse sizes. This approach enhances object detection performance in remote sensing images.

Fig. 5 illustrates the factors affecting the loss value. When the predicted and ground-truth boxes share the same aspect ratio but differ in width and height, AIoU provides more accurate loss values. The AIoU loss function, presented in (19), includes the original IoU loss, the d/c distance loss, and an optimized shape loss calculation. Equations (16) and (17) compute the absolute differences between the predicted and ground-truth boxes, normalized by their maximum values, addressing the issue of proportionally similar boxes. Equation (18) maps the shape penalty term to a range between 0 and 1, thereby improving both training speed and prediction accuracy

$$W_{\mathrm{dif}} = \frac{|\mathrm{BoxW}_{\mathrm{prd}} - \mathrm{BoxW}_{\mathrm{gt}}|}{\max\left(\mathrm{BoxW}_{\mathrm{prd}}, \mathrm{BoxW}_{\mathrm{gt}}\right)} \tag{16}$$

$$H_{\mathrm{dif}} = \frac{|\mathrm{BoxH}_{\mathrm{prd}} - \mathrm{BoxH}_{\mathrm{gt}}|}{\max\left(\mathrm{BoxH}_{\mathrm{prd}}, \mathrm{BoxH}_{\mathrm{gt}}\right)} \tag{17}$$

$$\mathcal{P}_{\mathrm{Shape}} = \left(1 - e^{-W_{\mathrm{dif}}}\right)^w + \left(1 - e^{-H_{\mathrm{dif}}}\right)^w \tag{18}$$

$$\mathcal{L}_{\mathrm{AIoU}} = 1 - \frac{B_{\mathrm{gt}} \cap B_{\mathrm{prd}}}{B_{\mathrm{gt}} \cup B_{\mathrm{prd}}} + \frac{d}{c} + \mathcal{P}_{\mathrm{Shape}} \tag{19}$$

where $\mathrm{BoxW}_{\mathrm{prd}}$ represents the width of the predicted box, $\mathrm{BoxW}_{\mathrm{gt}}$ the width of the annotated box, $\mathrm{BoxH}_{\mathrm{prd}}$ the height of the predicted box, and $\mathrm{BoxH}_{\mathrm{gt}}$ the height of the ground-truth box. In (18), w is a relatively small constant that controls the sensitivity and weighting of the shape difference terms W_{dif} and H_{dif}. Specifically, w acts as an exponent that adjusts the contribution of the width and height differences to the overall shape penalty $\mathcal{P}_{\mathrm{Shape}}$. A higher value of w increases sensitivity to larger differences, thereby amplifying the penalty for significant shape mismatches, whereas a lower w reduces sensitivity, making the penalty less severe for minor discrepancies. Thus, w serves as a tuning parameter to balance the influence of shape differences on the overall penalty, allowing the model to adapt to varying requirements for shape consistency.

IV. Experimental Results

We validate the proposed MFAE-YOLO through extensive experiments on three widely used aerial remote sensing datasets: NWPU VHR-10 [22], RSOD [55], and DIOR [56]. To demonstrate the effectiveness of our approach, we compare it against several state-of-the-art models.

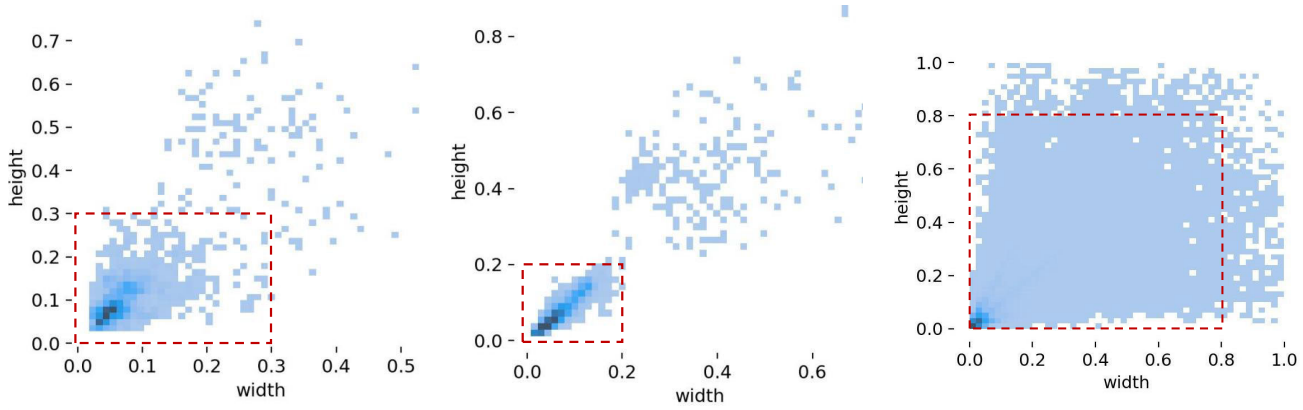
A. Experimental Dataset

High-quality datasets are crucial for training and testing object detection models [57]. Typically constructed using satellite and drone images with manual annotations, these datasets serve as essential benchmarks. For this study, we selected the NWPU VHR-10, RSOD, and DIOR datasets because of their representativeness and high-quality annotations, providing a comprehensive evaluation of the proposed detection models. Each dataset is carefully analyzed to assess its suitability for object detection in aerial remote sensing images.

NWPU VHR-10 is a standard benchmark for aerial object detection, comprising 800 ultrahigh-resolution images with horizontal bounding box (HBB) annotations across ten object categories: airplanes (ALs), ships, tanks, baseball fields, tennis courts (TCs), basketball courts, ground runways, harbors, bridges, and vehicles. It includes 3651 annotated objects, with 650 images annotated by experts to ensure high-quality labels. The objects vary in size, from large objects such as runways to smaller ones such as ALs and ships, making this dataset suitable for multiscale object detection. The dataset is split into training, validation, and test sets in a 6:2:2 ratio, with 390 training images, 130 validation images, and 130 test images.

RSOD comprises 936 images labeled in the PASCAL VOC format, covering four object categories: ALs, oil tanks, overpasses, and playgrounds. It includes 4993 aircraft instances across 446 images, 191 playground instances in 149 images, 180 overpasses in 176 images, and 1586 oil tanks in 165 images. This dataset exhibits a long-tail distribution, with a higher number of AL samples and fewer instances in other categories, adding complexity to the detection of less-represented classes. The images are divided into 560 training images, 188 validation images, and 188 test images, following a 6:2:2 split.

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(a) (b) (c)

Fig. 6. Distribution of object sizes across (a) NWPU VHR-10, (b) RSOD, and (c) DIOR.

TABLE I
EXPERIMENTAL PARAMETER SETTINGS

Parameter	Value
Epochs	400
Training Batch Size	64
Testing Batch Size	64
Image Size	640×640
Number of Worker Threads	8
Optimizer	AdamW
IflrAuto	True
Initial Learning Rate	0.01
Momentum	0.937
Patience	100
Random Seed for Reproducibility	0
Weight Decay	0.0005
w in $\mathcal{P}_{\text{Shape}}$ (AIOU loss)	4

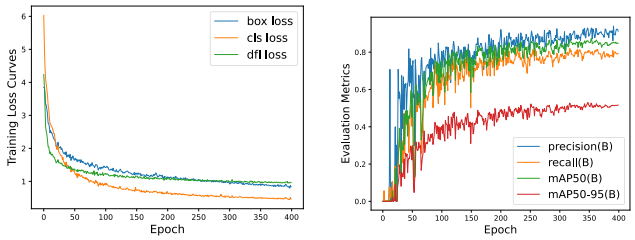
DIOR is a large-scale remote sensing benchmark containing 23 463 images with 192 472 object instances across 20 categories. Sourced from platforms such as Google Earth and Tianditu, it ensures diversity in geographic and scene variations. The objects vary significantly in size and appearance, presenting challenges such as high interclass similarity and intraclass diversity. The dataset is split into 5862 training images, 5863 validation images, and 11 738 test images.

As illustrated in Fig. 6, most objects in these datasets are small, with dimensions smaller than 20×20 pixels. NWPU VHR-10 exhibits irregular object sizes, RSOD primarily features square-shaped objects, and DIOR contains both larger and more diverse objects. These datasets provide comprehensive validation of the proposed method's effectiveness across a range of object sizes and shapes.

B. Implementation Details

Experiments are conducted on a server equipped with an Intel Xeon Platinum 8336C CPU (2.3 GHz) and two NVIDIA RTX A6000 GPUs (48 GB each), running Linux Ubuntu 5.15.0 (64-bit). The model is implemented using the PyTorch framework (Python 3.9.19, PyTorch 2.2.2). Key parameter settings are summarized in Table I.

To ensure fairness in comparing different models, we standardized the batch size to 64 for both training and testing.



(a) (b)

Fig. 7. Training indicators on NWPU VHR-10. (a) Training loss curves. (b) Evaluation metrics curves.

Since the learning rate and image segmentation size significantly influence object detection experiments, all the images from the three datasets were resized to a uniform resolution of 640×640 pixels. In addition, YOLO's dynamic learning rate adjustment mechanism was used to optimize training, with the momentum value set to 0.937 and the patience parameter set to 100. This means that if a model shows no improvement for 100 consecutive epochs during training, the process will automatically terminate, thereby saving training time and computational resources.

The evaluation follows standard object detection practices, with mean average precision (mAP) serving as the primary performance metric.

C. Comparative Experiment

Training results of MFAE-YOLO on the NWPU VHR-10 training set are shown in Fig. 7, demonstrating rapid loss reduction and stable convergence.

The proposed model exhibits consistent performance across datasets with varying object categories, successfully identifying both small-scale objects and larger sets spanning up to 20 categories. Compared with the existing YOLO series models, our method ranks first on all three datasets, excelling particularly on the large-scale DIOR, thereby validating its effectiveness on diverse, large-scale datasets. Our model demonstrates superior performance in both general object detection and remote sensing-specific applications. Compared with the latest YOLO-series models (e.g., YOLOv11), the

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TABLE II

COMPARISON TO OTHER REPRODUCED METHODS ON NWPU VHR-10. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY. † REPRESENTS REPRODUCED METHODS

Method	Venue	Airplane	Ship	ST	BD	TC	BC	GTF	Harbor	Bridge	Vehicle	mAP_{50}
Faster R-CNN [38]	TPAMI'17	99.7	85.5	99.2	95.9	88.2	92.0	99.7	92.1	43.3	86.6	88.3
YOLO-v4 [17] †	CVPR'21	99.4	92.6	**99.8**	95.2	98.3	75.2	94.1	93.1	71.9	87.8	90.7
YOLO-v5n [41] †	github'21	99.4	93.7	87.9	98.2	92.7	74.9	95.8	93.1	90.1	82.5	90.8
EGAT [58]	ICANN'21	97.3	96.7	97.1	96.5	86.6	94.4	94.1	86.1	80.1	90.7	92.0
ABNet [59]	TGRS'22	**100.0**	92.5	97.7	97.7	99.2	95.9	**99.8**	94.2	69.0	**95.6**	94.2
YOLO-v8n [42] †	github'23	99.4	96.3	89.5	**98.7**	95.1	76.6	95.3	92.6	88.7	88.2	92.0
YOLO-v9c [19] †	ECCV'24	98.0	90.2	89.5	98.5	92.6	84.9	96.6	93.8	67.3	91.2	90.3
YOLO-v11 [44] †	arXiv'24	99.1	92.5	92.4	97.6	94.9	78.1	96.6	96.3	77.4	90.1	91.5
LAR-YOLOv8 [48]	JSTARS'24	99.3	90.7	89.0	96.6	87.8	85.8	99.4	86.1	84.8	88.6	90.8
SOCS-BSS [60]	JSTARS'24	99.9	**97.3**	98.8	85.4	**100.0**	**97.7**	85.9	89.7	**96.6**	90.8	94.2
MFAE-YOLO (Ours)		99.5	97.1	94.6	98.2	95.8	91.4	95.6	**96.8**	88.0	90.0	**94.7**

TABLE III

COMPARISON TO OTHER REPRODUCED METHODS ON DIOR. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY. † REPRESENTS REPRODUCED METHODS

Method	AL	AT	BF	BC	B	C	D	ESA	ETS	GF	GB	HB	OP	S	SD	ST	TC	TS	V	WM	mAP_{50}
R-CNN [11]	35.6	43.0	53.8	62.3	15.6	53.7	33.7	50.2	33.5	50.1	49.3	39.5	30.9	9.1	60.8	18.0	54.0	36.1	9.1	16.4	37.7
Faster R-CNN [38]	53.6	49.3	**78.8**	66.2	28.0	70.9	62.3	69.0	56.0	68.9	56.9	50.2	50.1	27.7	**73.0**	39.8	75.2	38.6	23.6	45.4	54.1
CornerNet [61]	58.8	**84.2**	72.0	80.8	**46.4**	75.3	**64.3**	**81.6**	**76.3**	**79.5**	**79.5**	26.1	**60.6**	37.6	70.7	45.2	84.0	57.1	43.0	75.9	64.9
RetinaNet [53]	53.7	77.3	69.0	81.3	44.1	72.3	62.5	76.2	66.0	77.7	74.2	50.7	59.6	71.2	69.3	44.8	81.3	54.2	45.1	**83.4**	65.7
YOLO-v5n [41] †	64.7	66.2	77.3	84.6	31.8	74.0	45.4	59.3	52.5	66.5	66.3	61.2	52.6	88.6	65.7	70.7	86.8	50.0	44.4	73.5	64.7
LO-Det [62]	72.6	65.0	76.7	84.7	33.7	73.7	56.8	75.9	57.5	66.3	68.0	60.9	51.5	88.6	68.0	64.3	86.3	47.6	42.4	76.7	65.9
T-TRD-DA [63]	77.9	80.5	70.1	**86.3**	39.7	**77.9**	59.3	59.0	54.4	74.6	73.9	49.2	57.8	74.2	61.1	69.8	84.0	**58.8**	**50.5**	77.2	66.8
YOLO-v8n [42] †	76.7	71.5	76.5	85.2	33.7	75.6	53.3	60.3	53.6	74.0	65.0	**62.0**	54.8	88.8	62.2	70.1	87.1	51.1	45.5	74.1	66.1
YOLO-v10n [43] †	72.4	74.1	77.0	84.3	34.3	74.1	53.6	59.5	52.0	72.2	67.5	61.0	53.8	88.1	69.5	71.7	85.9	54.7	45.6	72.2	66.2
YOLO-v11 [44] †	76.2	67.4	76.9	85.9	27.9	73.6	47.8	59.3	51.2	72.7	63.9	51.7	50.1	87.4	60.8	67.0	86.1	45.5	43.1	71.5	63.3
MFAE-YOLO (Ours)	**80.9**	72.8	76.5	86.1	33.8	74.8	53.9	62.3	55.2	72.7	65.1	61.2	53.8	**89.4**	69.5	**72.6**	**87.4**	49.0	46.4	76.1	**67.0**

TABLE IV

COMPARISON TO OTHER REPRODUCED METHODS ON RSOD. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY. † REPRESENTS REPRODUCED METHODS

Method	Venue	Aircraft	Oil Tank	Overpass	Playground	mAP_{50}
Faster R-CNN [38]	TPAMI'17	70.8	90.1	78.7	98.0	84.4
Soft-NMS [64]	ICCV'17	76.1	90.3	81.3	98.8	86.6
SSAFNet [65]	IGARSS'20	95.8	98.4	84.7	92.5	92.8
YOLO-v5n [41] †	github'21	96.4	97.9	76.3	76.3	92.1
YOLO-v8n [42] †	github'23	95.5	97.2	81.9	97.9	93.1
Sig-NMS [66]	TGRS'24	80.6	90.6	**87.4**	99.1	89.4
YOLO-v9c [19] †	ECCV'24	**96.8**	97.1	66.0	98.2	89.5
YOLO-v11 [44] †	arXiv'24	96.0	97.5	74.1	96.4	91.0
LAR-YOLOv8 [48]	JSTARS'24	94.8	98.2	85.4	98.9	94.3
AGMF-Net [67]	TGRS'24	96.0	**99.0**	82.4	**99.7**	94.3
MFAE-YOLO (Ours)		**96.8**	97.9	85.0	99.3	**94.8**

proposed method achieves consistent improvements across multiple benchmarks: +3.2% mAP_{50} on NWPU VHR-10 (Table II), +3.7% mAP_{50} on DIOR (Table III), and +3.8% mAP_{50} on RSOD (Table IV). Furthermore, it outperforms state-of-the-art remote sensing detectors including ABNet [59], SOCS-BSS [60], LAR-YOLOv8 [48], LO-Det [62], T-TRD-DA [63], SSAFNet [65], and AGMF-Net [67], achieving the highest mAP_{50} scores on all three benchmark datasets.

For AL detection in Table II, which is a typical small object detection task, our method trails the top performer by only 0.5%. In detecting more challenging objects such as bridges, our model is only 8.6% behind the best model but surpasses the lowest ranked model by 44.7%. In Table III, our method performs best for small object detection tasks, such as ALs, ships (S), and storage tanks (STs), although it is 6.8% behind for overpasses (OP), yet still outperforms the weakest model by 22.9%. In Table IV, the model performs best for small targets such as aircraft but is 2.4% behind for overpass, still outperforming the weakest model by 19%. Across Tables II–IV, which involve large datasets with 20 categories, our method achieves above-average results in all the categories, demonstrating robust overall performance.

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TABLE V
COMPARISON OF COMPUTATIONAL EFFICIENCY. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Method	Venue	FPS	Params	GFLOPs
Faster R-CNN [38]	TPAMI'17	25.9	136.7M	401.7
RetinaNet [53]	ICCV'17	35.6	36.3M	163.5
YOLO-v8n [42]	github'23	**132.3**	**3.0M**	<u>8.1</u>
LAR-YOLOv8 [48]	JSTARS'24	55.6	26.4M	-
YOLO-v9c [19]	ECCV'24	75.3	25.3M	102.4
I-YOLO [46]	TVC'24	-	4.9M	18.1
M-YOLO [45]	JIFS'24	65.8	4.6M	10.1
EfficientDet [44]	CVPR'24	21.6	3.8M	**4.7**
MFAE-YOLO (Ours)		<u>107.1</u>	**3.0M**	11.9

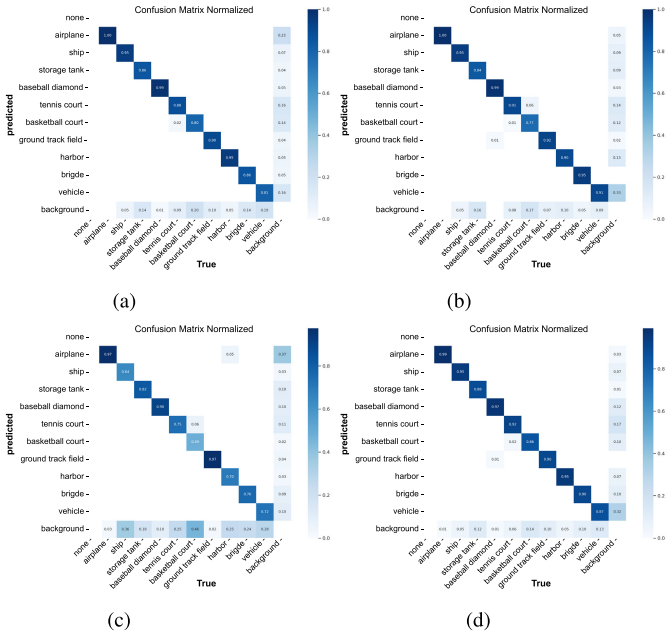


Fig. 8. Confusion matrices of different models on NWPU VHR-10. (a) YOLOv5. (b) YOLOv8. (c) YOLOv10. (d) MFAE-YOLO.

D. Performance Experiment

The consistent detection results across multiple remote sensing datasets underscore the robustness of our feature extraction method. To further assess the efficiency of various models, we conducted comprehensive performance evaluations on a remote sensing dataset. As illustrated in Table V, YOLOv8 achieved the highest frames per second (FPS) score, indicating superior real-time processing capabilities. Meanwhile, our proposed MFAE-YOLO model attained a competitive FPS of 107.1, demonstrating strong real-time performance. In terms of computational efficiency, EfficientDet [44] led in GFLOPS, whereas our model achieved a moderate score, suggesting room for improvement in computational complexity. Notably, MFAE-YOLO has the fewest parameters among the evaluated models, highlighting its efficiency in resource utilization. Overall, our model excels in scenarios prioritizing recognition accuracy over detection speed.

In addition, the confusion matrix comparisons in Fig. 8 show that MFAE-YOLO consistently outperforms YOLOv5, YOLOv8, and YOLOv10 on NWPU VHR-10. Notably, MFAE-YOLO exhibits higher detection accuracy for challenging objects such as basketball courts, which are typically affected by background interference.

E. Ablation Experiment

The MFAE-YOLO architecture comprises the GFFP module (which integrates the FCPS module), the C2F-FPEU module, and the AIoU loss function. We perform ablation experiments to assess the contribution of each component. To validate the multifeature attention mechanism proposed in this article, the ablation experiments were conducted following the image processing pipeline. First, the effectiveness of the GFFP module for global attention extraction in the backbone layer was verified. Next, the feature enhancement capability of the C2F-FPEU module in the neck layer was evaluated. Subsequently, the performance improvements of the proposed AIoU loss function were assessed. Finally, the combined effectiveness of all the modules was comprehensively validated.

As shown in Table VI, incorporating the GFFP module improves the baseline model's mAP_{50} by 1.9% on NWPU VHR-10. Adding the C2F-FPEU module results in a further increase of 0.4% mAP_{50}, and the AIoU loss function contributes an additional 0.4%. Although the baseline model already achieves a high mAP_{50} of 92%, these cumulative enhancements significantly improve overall performance. While the mAP values of certain categories may decrease due to varying feature extraction abilities of individual modules for targets of different sizes, experiments show that the overall mAP increases with the addition of our proposed modules. Moreover, subindicators in Table VI show significant improvements in detecting three small objects: aircraft, ships, and STs, demonstrating that the designed modules better identify small objects in complex remote sensing backgrounds and enhance overall recognition performance.

Tables VII and VIII further illustrate that the proposed modules significantly improve detection performance for small objects such as ALs and ships, as well as for larger objects such as sports venues. While the overall mAP_{50} improves with the inclusion of our modules, certain categories exhibit a decline in performance due to inherent challenges in those categories. For instance, the slight performance drop in the ground field (GF), groundbreaking (GB), and overpass (OP) categories may be attributed to the homogeneous textures, densely packed structures, or complex geometric patterns that are not adequately captured by the current modules. In contrast, high-performing categories such as ship (S), bridge (B), stadium (SD), ST, and TC consistently achieve high mAP_{50} scores. On the large-scale DIOR, most subindicators show stable improvement, proving the generalization ability of the proposed modules in object detection.

These results demonstrate that the proposed modules enhance remote sensing object detection performance across multiple datasets, with the degree of improvement varying based on dataset characteristics.

F. Error Analysis

Despite achieving strong recognition results, MFAE-YOLO has limitations. The introduction of a global

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TABLE VI

ABLATION EXPERIMENT ON NWPU VHR-10. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Modules				Airplane	Ship	ST	BD	TC	BC	GTF	Harbor	Bridge	Vehicle	mAP_{50}
GFFP	FPEU	AIoU	CIoU											
○	○	○	●	99.4	96.3	89.5	**98.7**	95.1	76.6	95.3	92.6	88.7	88.2	92.0
●	○	○	●	99.4	94.9	92.6	**98.7**	**96.6**	**95.9**	97.5	**98.3**	79.7	85.7	93.9 (+1.9)
●	●	○	●	98.7	91.7	93.8	98.2	94.7	86.2	**97.6**	95.1	**97.1**	**90.2**	94.3 (+0.4)
●	●	●	○	**99.5**	**97.1**	**94.6**	98.2	95.8	91.4	95.6	96.8	88.0	90.0	**94.7** (+0.4)

TABLE VII

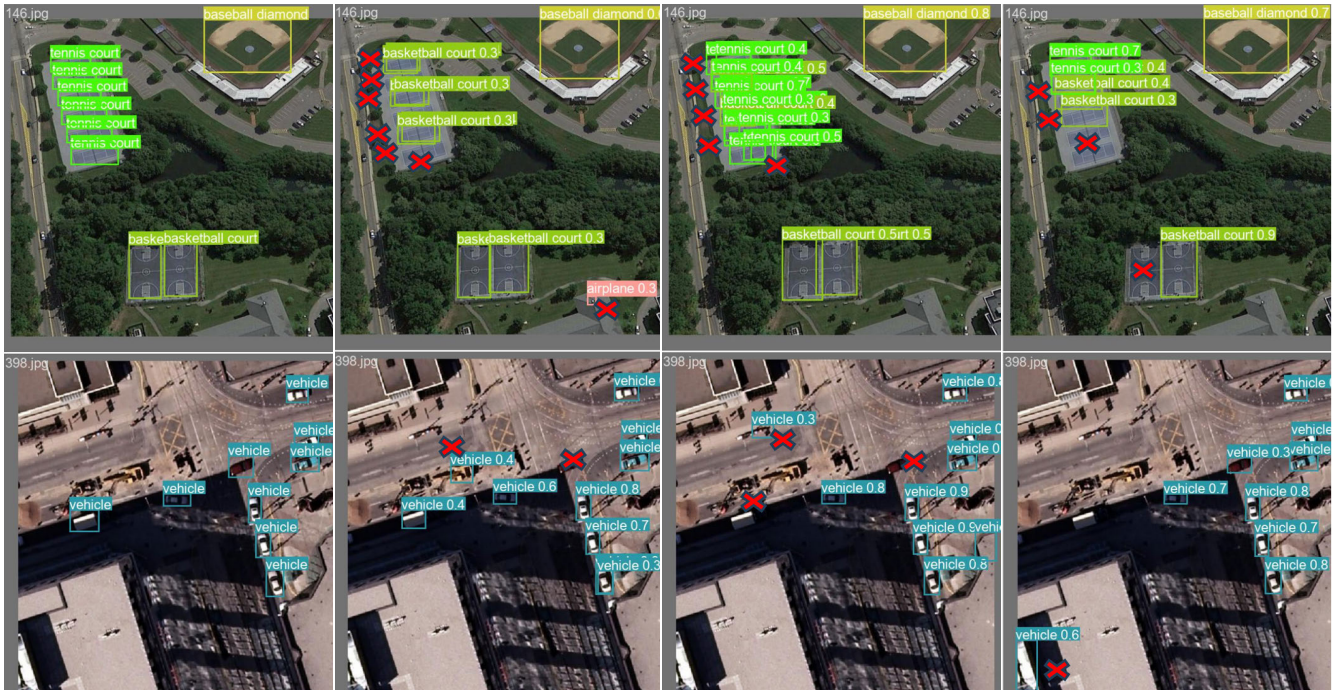
ABLATION EXPERIMENT ON DIOR. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Modules				AL	AT	BF	BC	B	C	D	ESA	ETS	GF	GB	HB	OP	S	SD	ST	TC	TS	V	WM	mAP_{50}
GFFP	FPEU	AIoU	CIoU																					
○	○	○	●	76.7	71.5	76.5	85.2	33.7	**75.6**	53.3	60.3	53.6	**74.0**	65.0	62.0	**54.8**	88.8	62.2	70.1	87.1	**51.1**	45.5	74.1	66.1
●	○	○	●	77.6	72.1	**78.3**	**86.6**	33.5	74.5	**54.7**	**62.8**	54.3	73.4	**67.8**	61.2	54.4	88.8	62.2	71.7	87.2	45.2	**46.7**	73.7	66.3 (+0.2)
●	●	○	●	79.5	**74.6**	76.9	86.1	33.7	74.6	51.5	61.0	53.6	73.7	67.0	**62.3**	53.1	89.1	67.5	72.4	86.9	48.4	45.8	74.3	66.6 (+0.3)
●	●	●	○	**80.9**	72.8	76.5	86.1	**33.8**	74.8	53.9	62.3	**55.2**	72.7	65.1	61.2	53.8	**89.4**	**69.5**	**72.6**	**87.4**	49.0	46.4	**76.1**	**67.0** (+0.1)

TABLE VIII

ABLATION EXPERIMENT ON RSOD. BOLD AND UNDERLINED NUMBERS HIGHLIGHT THE BEST AND SECOND-BEST RESULTS, RESPECTIVELY

Modules				Aircraft	Oil Tank	Overpass	Playground	mAP_{50}
GFFP	FPEU	AIoU	CIoU					
○	○	○	●	95.5	97.2	81.9	97.9	93.1
●	○	○	●	**96.9**	97.7	79.8	98.5	93.2 (+0.1)
●	●	○	●	96.7	**97.9**	84.7	**99.5**	94.7 (+1.5)
●	●	●	○	96.8	**97.9**	**85.0**	99.3	**94.8** (+0.1)

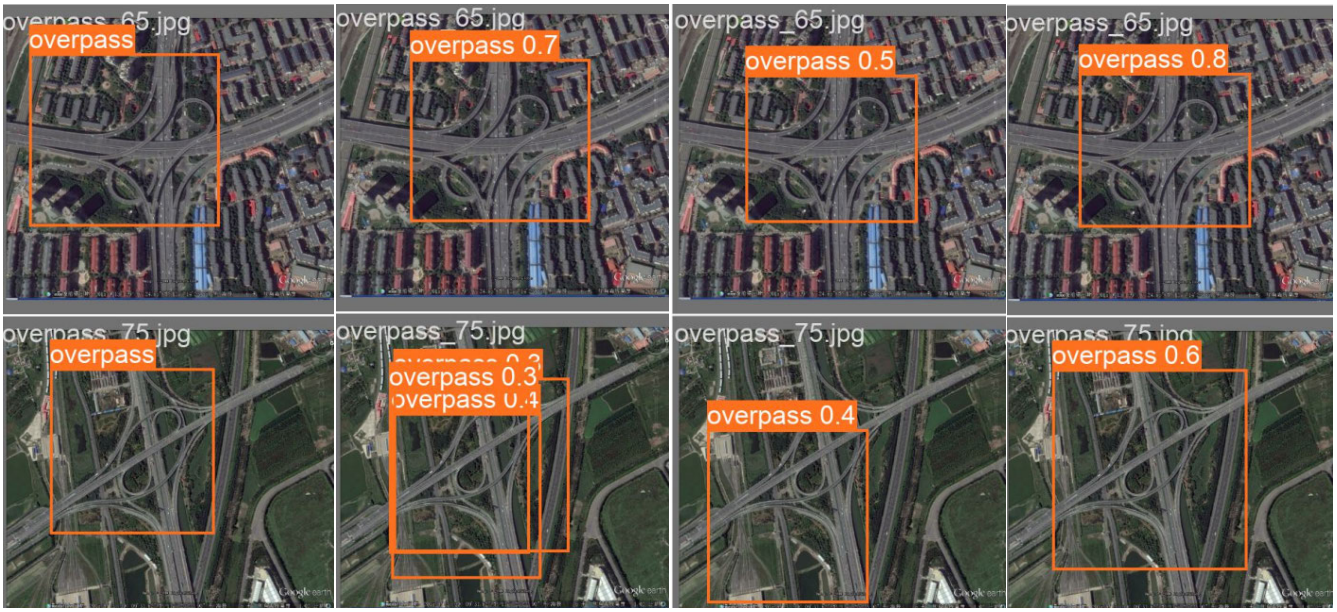


(a) (b) (c) (d)

Fig. 9. Detection performance comparison on NWPU VHR-10. (a) Annotation box. (b) YOLOv5. (c) YOLOv8. (d) MFAE-YOLO.

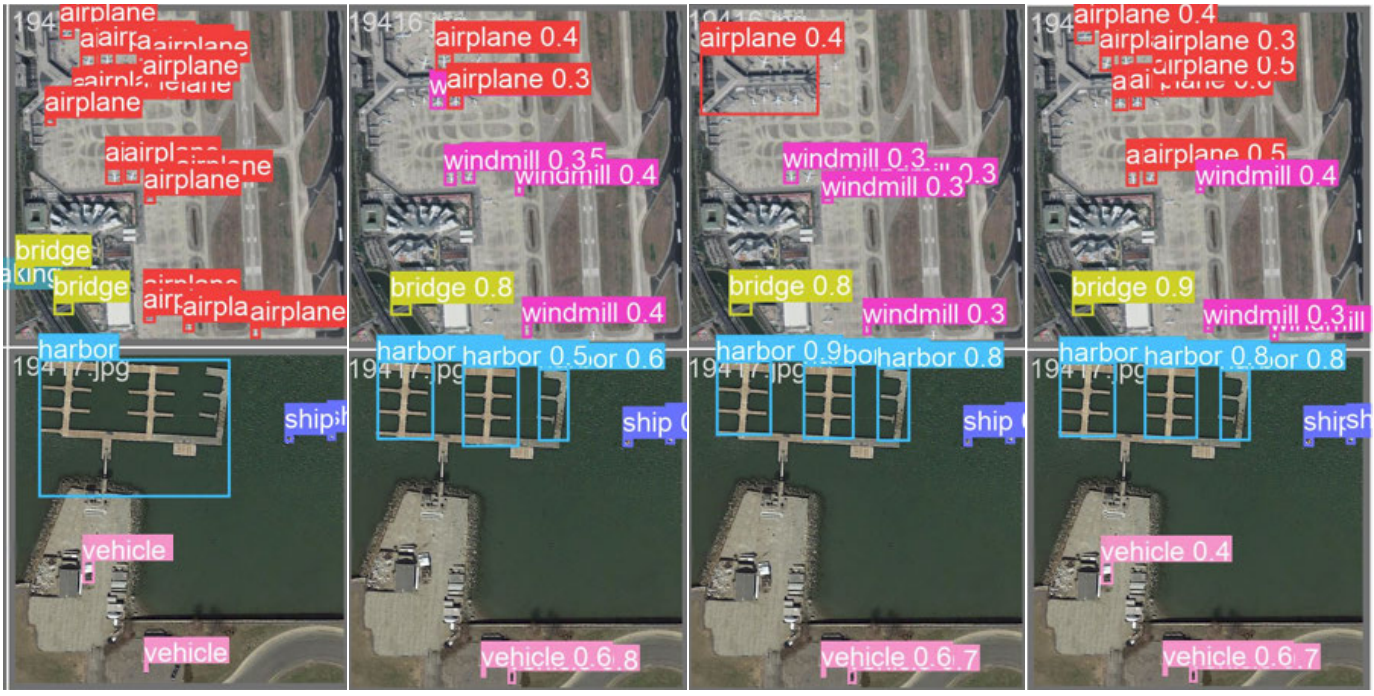
attention mechanism, multidimensional feature extraction, and AIoU enhances adaptability and detection capability, effectively reducing missed and false positives (as shown in Tables II–IV). However, adding modules increases the number of network parameters and computational complexity, which may reduce GFLOPs (as indicated in Table V) and hinder

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(a) (b) (c) (d)

Fig. 10. Detection performance comparison on RSOD. (a) Annotation box. (b) YOLOv5. (c) YOLOv8. (d) MFAE-YOLO.



(a) (b) (c) (d)

Fig. 11. Detection performance comparison on DIOR. (a) Annotation box. (b) YOLOv5. (c) YOLOv8. (d) MFAE-YOLO.

deployment on resource-constrained devices such as edge platforms. Moreover, our method has only been validated on visible light remote sensing images; further evaluation is needed for other modalities such as synthetic aperture radar (SAR) images. Future research will explore compressing the MFAE-YOLO model into an equally effective shallow network via model distillation techniques to address inference speed issues. In addition, we will focus on optimizing the loss function, exploring advanced neural network architectures, and developing more powerful modules to further improve object detection accuracy in remote sensing images.

G. Visualization Results

As illustrated in Fig. 9, MFAE-YOLO misses only one small vehicle, outperforming other models due to its advanced global feature extraction. In contrast, YOLOv5 and YOLOv8 miss several objects and produce false positives. In addition,

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MFAE-YOLO generates more accurate bounding boxes owing to its optimized AIoU loss function. For instance, in detecting irregular large objects such as basketball and baseball courts, MFAE-YOLO addresses the issue of repeated predictions and multiple bounding boxes compared to other models, thereby greatly enhancing detection effectiveness.

In Fig. 10, although all the methods correctly identify the overpass category, the challenge lies in accurately labeling its location. MFAE-YOLO provides the most precise annotations for challenging categories such as overpasses.

Furthermore, in Fig. 11, where images contain both small and large objects, MFAE-YOLO excels at detecting a variety of scales, including small aircraft and large bridges. Conversely, YOLOv5 and YOLOv8 frequently misclassify small aircraft as windmills, and for vehicle counting, our proposed method is the most accurate.

V. Conclusion

This article addresses the object detection challenge in aerial remote sensing images by introducing the MFAE-YOLO algorithm, an extension of YOLO designed for effective feature extraction and multiscale object detection. The MFAE-YOLO network integrates three plug-and-play modules, namely, the GFFP module, the FCPS module, and the C2F-FPEU module, along with enhancements to the YOLO backbone.

The proposed modules offer several improvements compared with the existing approaches. First, the GFFP module uses multiple attention mechanisms and incorporates residual connections to enhance global feature extraction at the initial input stage, ensuring stability. Second, the FCPS module combines SA, CA, and PA to enable multilevel extraction of image information. Finally, the C2F-FPEU module refines the original C2F module using a feature pooling extraction method specifically tailored to the characteristics of remote sensing images. In addition, we propose AIoU, an improved method for bounding box loss calculation. Comparative experiments on three aerial remote sensing datasets, NWPU VHR-10, RSOD, and DIOR, demonstrate the effectiveness of our approach, achieving mAP_{50} values of 94.7%, 94.8%, and 67.0%, respectively, which surpass those of all the baseline models. Ablation studies further validate the contributions of each module. We believe that our research will promote further advancements in remote sensing image analysis.

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